Lecture 23 24

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1 Range of Functions

1.1 Example 1: Rational Function Range

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \frac{2x}{1+x^2}$. Find the range $f(\mathbb{R})$.

1.1.1 Solution Method 1 (Inequalities)

We know $(x-1)^2 \geq 0 \implies x^2 - 2x + 1 \geq 0 \implies 2x \leq 1 + x^2$. Similarly, $(x+1)^2 \geq 0 \implies x^2+2x+1 \geq 0 \implies -2x \leq 1+x^2$. Thus, $-1 \leq \frac{2x}{1+x^2} \leq 1$. The range is contained in $[-1, 1]$.

1.1.2 Solution Method 2 (Inverting)

Let $y = \frac{2x}{1+x^2}$. We solve for x in terms of y.

$$2x = y(1 + x^2)$$

$$yx^2 - 2x + y = 0$$

Case 1: $y = 0$. Then $-2x = 0 \implies x = 0$. Case 2: $y \neq 0$. This is a quadratic equation in x. For x to be real, the discriminant must be non-negative:

$$D = (-2)^2 - 4(y)(y) = 4 - 4y^2 \geq 0$$

$$1 - y^2 \geq 0 \implies y^2 \leq 1 \implies y \in [-1, 1]$$

Solving for x:

$$x = \frac{2 \pm \sqrt{4 - 4y^2}}{2y} = \frac{1 \pm \sqrt{1 - y^2}}{y}$$

Answer: The range is $[-1, 1]$.

2 Injective Functions on $\mathbb{N}$

2.1 Example 2

Find all injective functions $f : \mathbb{N} \to \mathbb{N}$ such that $f(n) \leq n$ for all n.

2.1.1 Solution

- For $n = 1$: $f(1) \leq 1 \implies f(1) = 1$.
- For $n = 2$: $f(2) \leq 2 \implies f(2) \in \{1, 2\}$. Since f is injective and $f(1) = 1$, we must have $f(2) = 2$.
- For $n = 3$: $f(3) \in \{1, 2, 3\}$. Since $f(1) = 1, f(2) = 2$, we must have $f(3) = 3$.

Hypothesis: $f(n) = n$ for all n. **Proof by Induction:**

1. **Base Case:** $f(1) = 1$ (Done).
2. **Step:** Suppose $f(k) = k$ for $k = 1, 2, \ldots, n$.
3. Consider $f(n + 1)$. We know $f(n + 1) \leq n + 1$, so $f(n + 1) \in \{1, 2, \ldots, n, n + 1\}$. Since f is injective and the values $1, \ldots, n$ are already taken by $f(1), \ldots, f(n)$, the only remaining option is $f(n+1) = n + 1$.

3 Bijections

3.1 Example 3: Quadratic Bijection

Prove $f : [3, \infty) \to [5, \infty)$ defined by $f(x) = x^2 - 6x + 14$ is a bijection.

3.1.1 Solution

Let $y \in [5, \infty)$. We solve $x^2 - 6x + 14 = y$.

$$x^2 - 6x + 9 = y - 5$$
$$(x - 3)^2 = y - 5$$
$$x - 3 = \pm\sqrt{y - 5}$$
$$x = 3 \pm \sqrt{y - 5}$$

Since the domain is $[3, \infty)$, we require $x \geq 3$. Thus, we must choose the positive root: $x = 3 + \sqrt{y - 5}$. Since $y \geq 5$, $y - 5 \geq 0$, so the square root exists. This shows for every y, there is a unique x, so f is a bijection.

3.2 Example 4: Modular Arithmetic

Let $f : \mathbb{Z}_4 \to \mathbb{Z}_4$ defined by $f(\bar{x}) = \overline{3x+1}$. Prove f is bijective.

3.2.1 Solution

We evaluate f for all elements in $\mathbb{Z}_4 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$:

- $f(\bar{0}) = \bar{1}$
- $f(\bar{1}) = \overline{3(1)+1} = \bar{4} = \bar{0}$
- $f(\bar{2}) = \overline{3(2)+1} = \bar{7} = \bar{3}$
- $f(\bar{3}) = \overline{3(3)+1} = \overline{10} = \bar{2}$

Since the range is $\{\bar{1}, \bar{0}, \bar{3}, \bar{2}\}$, which is all of $\mathbb{Z}_4$, f is bijective.

3.3 Example 5: Circle Segment

Let $f : [0,1] \to [0,1]$ defined by $f(x) = \sqrt{1-x^2}$. Prove it is a bijection and find f^{-1}.

3.3.1 Solution

Let $y = \sqrt{1-x^2}$. Since range is $[0,1]$, $y \geq 0$.

$$y^2 = 1 - x^2$$

$$x^2 = 1 - y^2$$

$$x = \pm\sqrt{1-y^2}$$

Since domain is $[0,1]$, $x \geq 0$, so $x = \sqrt{1-y^2}$. Thus, the inverse function is $f^{-1}(y) = \sqrt{1-y^2}$. Note that $f = f^{-1}$.

4 Permutations

4.1 Counting Bijections

How many bijections are there from a set of n elements to itself? **Answer:** $n!$ (n factorial).

4.2 Composition and Inverse of Permutations

Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}$.

- Find α^{-1}: Swap rows and sort.

$$\alpha^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

  Check: $\alpha \circ \alpha^{-1} = id$.

4.3 Example 6: Solving Permutation Equations

Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 1 & 3 \end{pmatrix}$. Find the permutation x such that $\alpha \circ x = \beta$.

4.3.1 Solution

Multiply by α^{-1} on the left:

$$x = \alpha^{-1} \circ \beta$$

First, find α^{-1}:

$$\alpha^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 2 & 3 \end{pmatrix}$$

Now compute composition (apply β first, then α^{-1}):

- $1 \xrightarrow{\beta} 4 \xrightarrow{\alpha^{-1}} 2$
- $2 \xrightarrow{\beta} 2 \xrightarrow{\alpha^{-1}} 5$
- $3 \xrightarrow{\beta} 5 \xrightarrow{\alpha^{-1}} 3$
- $4 \xrightarrow{\beta} 1 \xrightarrow{\alpha^{-1}} 4$
- $5 \xrightarrow{\beta} 3 \xrightarrow{\alpha^{-1}} 1$

$$x = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 3 & 4 & 1 \end{pmatrix}$$

5 Piecewise Functions and Inverses

5.1 Example 7

Let $f : \mathbb{R} \to \mathbb{R}$ defined by:

$$f(x) = \begin{cases} \sqrt{x-1} & \text{if } x \geq 1 \\ \frac{1}{x-1} & \text{if } x < 1 \end{cases}$$

Find $f^{-1}(y)$.

5.1.1 Solution

Analyze the range of each piece:

1. For $x \geq 1$: $y = \sqrt{x-1} \geq 0$.

$$y^2 = x - 1 \implies x = y^2 + 1$$

2. For $x < 1$: $x - 1 < 0 \implies y = \frac{1}{x-1} < 0$.

$$x - 1 = \frac{1}{y} \implies x = \frac{1}{y} + 1$$

Combining these inverses based on the value of y:

$$f^{-1}(y) = \begin{cases} y^2 + 1 & \text{if } y \geq 0 \\ \frac{1}{y} + 1 & \text{if } y < 0 \end{cases}$$

6 Classification of Functions

6.1 Symmetry (Even and Odd Functions)

Let $D \subset \mathbb{R}$ be symmetric around 0 ($x \in D \iff -x \in D$).

- **Odd Function:** $f(-x) = -f(x)$. (Graph symmetric about origin). Example: $f(x) = x^3$, $f(x) = \sin x$.
- **Even Function:** $f(-x) = f(x)$. (Graph symmetric about y-axis). Example: $f(x) = x^2$, $f(x) = \cos x$.

6.2 Monotonicity

- **Increasing:** $x_2 \geq x_1 \implies f(x_2) > f(x_1)$. (Strictly: $>$).
- **Non-decreasing:** $x_2 > x_1 \implies f(x_2) \geq f(x_1)$.

6.3 Periodicity

- **Periodic:** $\exists T > 0$ such that $f(x+T) = f(x)$ for all x.
- The smallest such T is the **fundamental period**.

7 Specific Function Types

1. **Linear/Affine:** $f(x) = ax + b$.
   - Odd if $b = 0$.
2. **Quadratic:** $f(x) = ax^2 + bx + c$.
   - Even iff $b = 0$ (Vertex on y-axis).
3. **Cubic:** $f(x) = ax^3 + bx^2 + cx + d$.
4. **Power Function:** $f(x) = x^n$ $(n \in \mathbb{N})$.
5. **Polynomial:** $f(x) = a_n x^n + \cdots + a_0$.
6. **Rational:** $f(x) = \frac{P(x)}{Q(x)}$.

8 Rational Power Functions and Inverse

Let $f(x) = x^n$ for $x \in [0, \infty)$. The inverse is $g(x) = x^{1/n} = \sqrt[n]{x}$.

- Graphs of f and f^{-1} are symmetric about $y = x$.

8.1 Rational Exponents

$$f(x) = x^{m/n} = (x^{1/n})^m = (\sqrt[n]{x})^m$$

8.2 Irrational Exponents

How to define $x^{\sqrt{2}}$? We approximate $\sqrt{2}$ with a sequence of rationals: $1.4, 1.41, 1.414, \ldots$ Then $x^{\sqrt{2}}$ is the limit of the sequence $x^{1.4}, x^{1.41}, \ldots$.